

Final Assignment

Statistical Programming & Machine Learning

Assignment Guidelines

Deadline: 08/12/2025 23:59

Earlier submissions are encouraged!

You are expected to submit:

1. A Word document:
 - Clearly explaining and visualizing your answers to the practical questions,
 - Any questions indicated as 'Word Document' must be solved by hand, with the calculations and reasoning explained in the Word document, unless otherwise stated,
 - Write the full names of all the group members at the top of the document.
2. A single iPython notebook:
 - Containing all your code for the assignment,
 - Write the full names of all the group members at the top using Markdown,
 - Clearly mark your answer for each question (e.g. Q1.A.2),
 - Use markdowns where necessary,
 - Use comments on your code where markdowns are not sufficient.

Requirements:

- For each answer in the Word document, and for each code segment in the notebook:
 - Indicate the name of the primary contributor (the group member who done the most work for that question) next to the question or code segment header.
- General Requirements:
 - If you use any external resources, you must reference them properly and refrain from **plagiarism**,
 - Each group can consist of up to **three people**, but no more,
 - It is enough for just one person from the group to submit,
 - This assignment, combined with Midterm assignment, is a requirement for passing (**G/VG**),
 - You can leverage **AI** but do not put your AI outputs as your solutions, your solutions must be unique and your own work,
 - If you use any code to visualize something for the practical questions, make sure you mark it clearly in both the code and Word submissions,
 - Clearly mark this segment in your code,
 - Clearly mark this segment in your word document.
 - Present all steps, explanations, and code in an organized and transparent manner,
 - Clearly show your work.

Good Luck and Have Fun!

Question 1: Price and Speed of Sale Prediction

Using the provided “Used Car Sales” dataset, you will fit two models: a Linear Regression model for price prediction and a Logistic Regression model for predicting a quick sale.

A. Linear Regression: Price Prediction

1. **Model Fitting (Code Submission):** Fit a Linear Regression model using Engine Size and Mileage to predict Price (€).
2. **Visualization (Word Document):** Generate a 3D scatter plot of the data points and the fitted regression plane.

Tip: Your plot should have Engine Size on one horizontal axis, Mileage on the other, and Price on the vertical axis. The plane represents your model's predictions.

3. **Analysis (Word Document):** Based only on the visualization, describe the relationship between each feature (Engine Size and Mileage) and the Price.

B. Logistic Regression: Quick Sale Classification

1. **Model Fitting (Code Submission):** Fit a Logistic Regression model using Engine Size and Mileage to predict Sold in 7 Days? (Y/N).
2. **Visualization (Word Document):** Generate a 2D scatter plot showing the data points for 'Sold in 7 Days?'. Colour the points based on their actual class (Y/N). Overlap this with the decision boundary generated by your model.

Tip: Plot Mileage on the X-axis and Engine Size on the Y-axis. The decision boundary is the line where the model's predicted probability is 0.5.

3. **Analysis (Word Document):** Explain, based on the decision boundary visualization, which used car sales the model predicts will be sold in 7 days and which won't. (e.g., "cars to the left/right of the line are predicted to be sold/not"). Identify and list any data points (Car No.) that appear to be misclassified by the decision boundary shown in your plot.

Question 2: The Logic of Failure

A. The "Failure Pathway" Decision Tree

You are tasked with using a single Decision Tree to create simple rules for management. Your company wants you to find the clearest "pathway to failure" in the provided "Software Project Success" dataset.

1. **Model Fitting (Code Submission):** Fit a Decision Tree Classifier to the data (predicting Success?).
2. **Visualization & Path Extraction (Code Submission & Word Document):**
 - a) Visualize the Tree: Plot the full Decision Tree.
 - b) Identify the Path: Trace and write down the shortest sequence of conditions (splits) that leads to a leaf node predicting the highest proportion of 'N' (Failure) projects. This is your *Failure Pathway*.
3. **Analysis (Word Document):** Based on the *Failure Pathway* you identified, formulate a single, easy-to-read Project Management Rule (e.g., "If **X** is less than **A** AND **Y** is greater than **B**, then failure is highly likely.").

B. The "Black Box" Consensus

For your second task, use a Random Forest providing a more robust measure of which features are globally important, even if its individual decisions are complex.

1. **Model Fitting (Code Submission):** Fit a Random Forest Classifier to the data.
2. **Feature Importance (Code Submission):** Extract the Feature Importances from your RF model.
3. **Visualization & Interpretation (Word Document):** Create a bar chart of the RF Feature Importances.
4. **Hypothesis & Defence (Word Document):**
 - a) Hypothesis: The Decision Tree's "Failure Pathway" (from part A) relies on a particular feature. Is this feature also ranked as the most important by the Random Forest? (Yes/No answer required).
 - b) Defence: If the feature is not ranked most important by the RF, explain how a feature can be crucial for the single tree's specific failure path but only moderately important for the Random Forest's overall classification ability. Hint: Consider how the RF handles redundant or correlated features.

Question 3 - The K-Means Balancing Act (Word Document)

A small specialty coffee roaster is analysing new bean samples based on two metrics: Aroma Score and Acidity Level. They want to cluster the beans into two groups ($K=2$) to guide their roasting profiles.

A. Initial Setup

1. **Visualization (by hand):** Plot the six data points (P1 to P6) on a 2D graph.
2. **Initial Centroid Selection:** Set the initial centroids (C_A and C_B) to be the first and last data points:
 - Centroid A (C_A): (1, 9)
 - Centroid B (C_B): (9, 1)

B. Iteration 1

1. **Assignment Step:** For each of the six data points (P1 to P6), calculate the Euclidean distance to both C_A and C_B .
2. **Show Work:** Present a simple table showing the distances and the resulting cluster assignment (A or B) for all six points.
3. **Update Step:** Calculate the coordinates of the new centroids ($C_{A,new}$ AND $C_{B,new}$) based on the means of the points assigned to each cluster.

C. Iteration 2

1. **Assignment Step:** Using the new centroids from the end of Iteration 1, repeat the process: calculate the distance from all six points to the new C_A and C_B .
2. **Show Work:** Present the assignment table and the resulting cluster assignment.
3. **Update Step:** Calculate the coordinates of the centroids for the end of Iteration 2.
4. **Convergence Analysis:** Based on your two iterations, explain whether the algorithm has converged or if a third iteration would be required. (Convergence occurs when no point changes its cluster assignment in an iteration).

D. Final Visualization

1. **Final Plot (by hand):** On your initial plot from part A, visually mark the final cluster assignments (e.g., circle points in Cluster A, square points in Cluster B and star final C_A and C_B).
2. **Interpretation:** What two distinct types of coffee beans (based on Aroma and Acidity) do the final clusters represent?

Question 4 - Uncorking the Data (Code Submission)

For this final programming challenge, we will use the Wine recognition dataset built into scikit-learn (`sklearn.datasets.load_wine`).

The Data: This dataset is the result of a chemical analysis of wines grown in the same region in Italy. The analysis determined the quantities of 13 constituents found in each of the three types of wines.

- **Features:** 13 variables including Alcohol, Malic acid, Ash, Alcalinity of ash, Magnesium, Total phenols, Flavanoids, Nonflavanoid phenols, Proanthocyanins, Color intensity, Hue, OD280/OD315 of diluted wines, and Proline.
- **The Unknown:** We have a collection of samples, but for the first part of this task, we will pretend we do not know how many distinct types of wine are present or which sample belongs to which type.

Part A: Unsupervised Exploration (The Hunt for Structure)

Hint: K-Means calculates distances between points. Consider if 'Magnesium' (values ~100) and 'Nonflavanoid phenols' (values ~0.3) should be treated equally without adjustment.

1. High-Dimensional Analysis (The Elbow):

- a) Using the full 13-feature dataset, perform the Elbow Method to determine the optimal number of clusters (K).
- b) Plot the Inertia (Sum of Squared Errors) for K=1 to K=10.
- c) Identify the "elbow" of the curve and state your choice for the optimal K.

2. Dimensionality Reduction (PCA):

- a) Apply Principal Component Analysis (PCA) to compress the dataset into 2 Principal Components.

3. Low-Dimensional Analysis (Silhouette):

- a) Using only the new 2-component PCA data, determine the optimal number of clusters (K) using the Silhouette Score.
- b) Calculate and print the Silhouette Score for K=1 to K=10.
- c) Does the Silhouette Score suggest the same K as the Elbow method?

4. Final Unsupervised Visualization:

- a) Fit a K-Means model on the PCA data using the best K you found in step 3.
- b) Generate a scatter plot: X-axis = PC1, Y-axis = PC2.
- c) Colour the points based on the Cluster Labels assigned by K-Means.

Part B: Supervised Classification (The Face-Off)

The Reveal: It turns out there are exactly 3 distinct Cultivars (classes 0, 1, and 2) in this dataset. Your goal now is to build a model that can predict the Cultivar of a wine based on its chemical profile.

1. Data Setup:

- a) Reload the dataset or use the variables to access the actual target labels (y).
- b) Split the data (original 13 features) into a Training set and a Test set (80:20 split).

2. Model Selection: *You will train and compare two different classification models.*

- a) **Model 1 (The "Classic"):** Choose a model we have covered extensively in class that is well-suited for multi-class classification (e.g., Random Forest or Logistic Regression).
- b) **Model 2 (The "Wildcard"):** Choose a model from one we have covered in class, or a model we have not explicitly covered or focused on, well-suited for multi-class classification.

3. Visual Comparison:

- a) Train both models on the training set.
- b) Evaluate them on the test set.
- c) Create two side-by-side Confusion Matrices (using heatmaps) to visually compare their performance.

4. Defence:

- a) Which model performed better?
- b) Defend your choice for "Model 2" (The Wildcard). Why did you think it would be a good fit for this specific dataset?

Part C: The Tri-View Reality Check

You have an unsupervised view (mathematical similarity), a supervised view (learned patterns), and the ground truth (reality). We want to visualize how these three perspectives differ.

1. Data Preparation for Visualization

- a) To make the plots comparable, we need a common 2D representation. Use the **PCA-transformed** data (from Part A) as your X and Y coordinates for **all points**.
- b) For the **Supervised Model Predictions**: run your best model from Part B on the full dataset (or the Test set if you prefer but clearly label it) to generate a list of predicted classes.
- c) For the **Cluster Model**: use the labels from your best K-Means run in Part A.

2. The Side-by-Side Visualization

- a) Create a single figure with 3 Subplots arranged in a row (1x3).
- b) Subplot 1 - Unsupervised: Scatter plot of PCA data, coloured by K-Means Cluster Labels. Title: "Mathematical Clusters".
- c) Subplot 2 - Supervised: Scatter plot of PCA data, coloured by Best Model Predictions. Title: "Model Predictions".
- d) Subplot 3 - Ground Truth: Scatter plot of PCA data, coloured by Actual Labels (y). Title: "Actual Cultivars".

3. Analysis & Defence: *Write a brief analysis comparing these three views.*

Address the following:

- a) **Defend the Unsupervised Result:** Look at Subplot 1 vs. Subplot 3. Where did K-Means fail to capture the reality? Explain why simple distance-based clustering might have failed there (e.g., "The clusters in reality are not spherical..." or "Class 2 is spread out, causing K-Means to split it").
- b) **Defend the Supervised Result:** Look at Subplot 2 vs. Subplot 3. Did your Supervised model capture the edge cases better than K-Means? Why? (e.g., "The SVM decision boundary was able to separate the slightly overlapping points that K-Means grouped together").
- c) **The "Ambiguity" Check:** Identify a specific region on the plots where the Clusters and the Supervised Predictions disagree. Which one was closer to the Ground Truth?